Argand Diagrams

Chapter Overview

Complex numbers can be represented as points in a plane.

1. Plot a number using its real and imaginary parts.
2. Describe it using a distance and an angle.
3. Multiply and divide using modulus–argument form.
4. Interpret equations and inequalities as loci and regions.

Angles are in radians unless stated otherwise.

Argand Diagrams 3

Plot $1+i,\ 2-3i,\ -1+2i,\ -2i$.

For $z = x + iy$, the horizontal coordinate is $\operatorname{Re} z = x$, and the vertical coordinate is $\operatorname{Im} z = y$.

Sketch

But why visualise complex numbers? 4

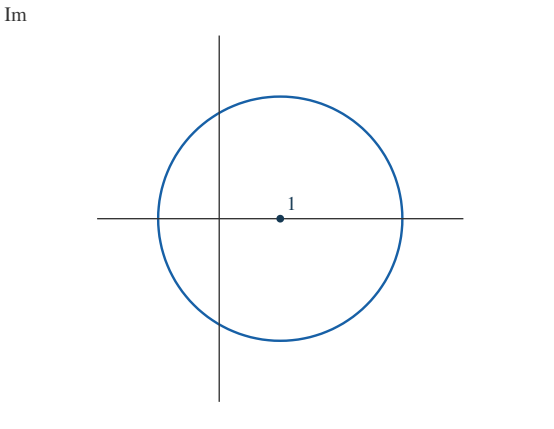
Geometry behind algebra

For a polynomial with real coefficients, non-real roots occur in conjugate pairs. On an Argand diagram they reflect in the real axis.

The four fourth roots of $1+i$ lie on a circle and are separated by angles of $\pi/2$.

The equation $|z-1| = 2$ describes a circle with centre $(1, 0)$ and radius 2.

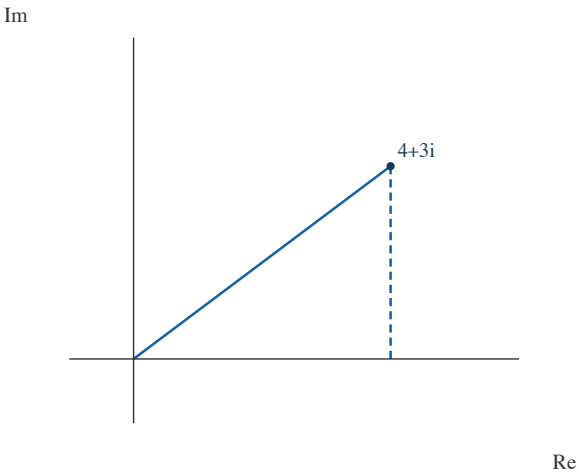
Complex-number iteration also generates geometric patterns such as the Mandelbrot set.



Modulus and Argument 5

For $z = 4 + 3i$, find its modulus r and argument θ.

Modulus and argument



The principal argument satisfies $-\pi < \arg z \leq \pi$. The argument of zero is undefined.

In general, $|x + iy| = \sqrt{x^2 + y^2}$. Use the quadrant to choose the correct argument.

Examples 6

1. Find the modulus and principal argument of

   $5 + 12i$

1

2. Find the modulus and principal argument of

   $-1 + i$

2

3. Find the modulus and principal argument of

   $-2i$

3

4. Find the modulus and principal argument of

   $-1 - 3i$

4

Test Your Understanding 7

Given $z = 2 - 3i$:

a. Find z^2.

b. Find $|z^2|$.

c. Find $\arg(z^2)$.

d. Plot z and z^2.

a

b

c

Sketch

Exercise 2B Pearson Core Pure 1, p. 21-23 8

3. Plot $z = -40 - 9i$ and find its principal argument.

3

Sketch

4. Given $z = 3 + 4i$, find z^2, its modulus and argument, and plot both.

4

Sketch

5. Let $z_1 = 4 + 6i$, $z_2 = 1 + i$. Find z_1/z_2, its modulus and argument.

5

7. Let $z = 26/(2 - 3i)$. Find z, z^2, $|z|$ and $\arg(z^2)$.

7

8. Let $z_1 = 4 + 2i$, $z_2 = 2 + 4i$, $z_3 = a + bi$.

a. Find $|z_1 + z_2|$.

b. Express $w = z_1 z_3/z_2$ in terms of a, b.

c. Given $w = 21/5 - 22i/5$, find a, b.

d. Find $\arg w$.

a

b

c

d

Exercise 2B Pearson Core Pure 1, p. 21-23 11

9. Let $w = 6 + 3i$. Find its modulus and argument. If $\arg(\lambda + 5i + w) = \pi/4$, find real λ.

9

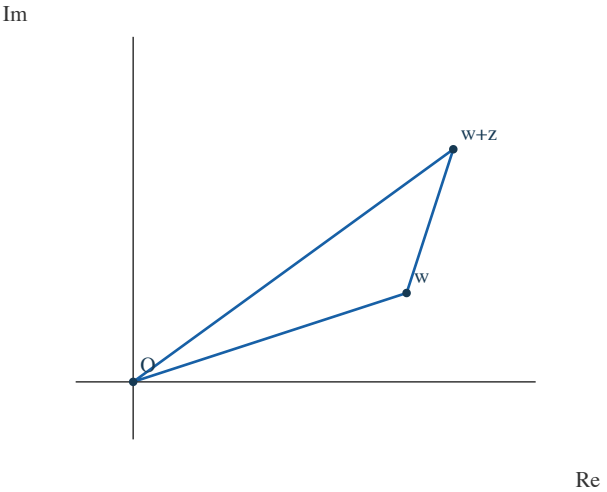
11. Let $w = k + i$, $z = -4 + 5ki$. Given $\arg(w + z) = 2\pi/3$, find real k.

11

Exercise 2B Pearson Core Pure 1, p. 21-23 12

12. Given $|w| = 5$, $\arg w = \pi/10$, $\arg z = 2\pi/5$, and $\arg(w + z) = \pi/5$, find $|z|$.

Modulus



Modulus-Argument Form 13

If $z = x + iy$, with modulus r and argument θ, then

$x = r\cos\theta, \qquad y = r\sin\theta$

Modulus–argument form

Use the full quadrant information when finding θ.

Angles differing by 2π represent the same direction.

The principal argument is chosen from $(-\pi, \pi]$.

Example 14

1. Express $-\sqrt{3} + i$ in modulus–argument form.

1

2. Express $-1 - \sqrt{3}i$ in modulus–argument form.

2

Initial questions 15

1. Convert to modulus–argument form.
   a. $2 + 2i$
   b. $3i$
   c. $-3 + 4i$
   d. $1 - \sqrt{3}i$
   e. $-2 - 5i$
   f. -20
   g. $7 - 24i$
   h. $-5 + 5i$

2. Convert to modulus–argument form.
   a. $3/(1 + i\sqrt{3})$
   b. $1/(2 - i)$
   c. $(1 + i)/(1 - i)$

3. Convert to Cartesian form, where $\operatorname{cis}\theta = \cos\theta + i\sin\theta$.
   a. $5\operatorname{cis}(\pi/2)$
   b. $\frac{1}{2}\operatorname{cis}(\pi/6)$
   c. $6\operatorname{cis}(5\pi/6)$
   d. $3\operatorname{cis}(-2\pi/3)$
   e. $2\sqrt{2}\operatorname{cis}(-\pi/4)$
   f. $-4\operatorname{cis}(7\pi/6)$

Solutions 16

Exercise answers: 1

Each pair is (r, θ).

Exercise answers: 2

Exercise answers: 3

Exercise 2C Pearson Core Pure 1, p. 24 — 17

4. Write $4\,\mathrm{cis}(2\pi/3)$ in Cartesian form and plot it.

4

Sketch

5. Find the number with modulus 7 and argument $11\pi/6$.

5

6. Find the number with modulus 5 and argument $-4\pi/3$.

6

Multiplying and Dividing Complex Numbers — 18

For $z_1 = r_1\,\mathrm{cis}\,\theta_1$, $z_2 = r_2\,\mathrm{cis}\,\theta_2$:

Product

Quotient

The addition formulae for sine and cosine give

$$(\cos\alpha + i\sin\alpha)(\cos\beta + i\sin\beta)$$
$$= \cos(\alpha+\beta) + i\sin(\alpha+\beta)$$

Reduce the resulting argument modulo 2π when a principal argument is requested.

Example

Multiplying and Dividing Complex Numbers 19

Simplify in Cartesian form:

$$\frac{\sqrt{2}\,\mathrm{cis}(\pi/12)}{2\,\mathrm{cis}(5\pi/6)}$$

Answer

Exercise 2D Pearson Core Pure 1, p. 27-28 20

4a. Simplify $\mathrm{cis}(5\theta)/\,\mathrm{cis}(2\theta)$.

a

4b. Simplify $\sqrt{2}\,\mathrm{cis}(\pi/2)/(\frac{1}{2}\,\mathrm{cis}(\pi/4))$.

b

4c. Simplify $3\,\mathrm{cis}(\pi/3)/(4\,\mathrm{cis}(5\pi/6))$.

c

4d. Simplify $(\cos 2\theta - i\sin 2\theta)/\,\mathrm{cis}(3\theta)$.

d

Exercise 2D Pearson Core Pure 1, p. 27-28 — 21

5. Let $z = -9 + 3\sqrt{3}i$ and $w = \sqrt{3}\,\mathrm{cis}(7\pi/12)$.

a. Express z in modulus–argument form.

b. Find zw and z/w in that form.

a

b

Loci — 22

What is the locus $|z| = 3$?

Locus

Sketch

A quick reminder... 23

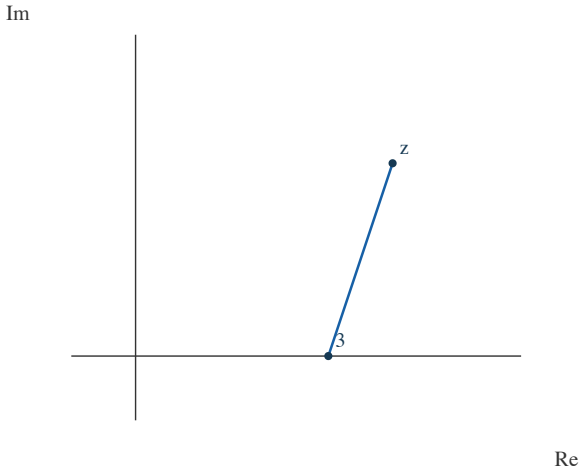
Let $z = x + iy$.

Distance from origin

Distance from (3,0)

In general, $|z - a - bi|$ is the distance from (a, b).

Therefore $|z - a - bi| = r$, with $r > 0$, is a circle.



Circular loci 24

Sketch $|z - 5 - 3i| = 3$ and give its Cartesian equation.

Circle

Sketch

Perpendicular bisector loci 25

Sketch $|z| = |z - 6i|$.

Locus

Sketch

Test Your Understanding So Far 26

Find the locus $|z - 3| = |z + i|$. Where does it meet $\operatorname{Re} z = 0$?

Equation

Intersection

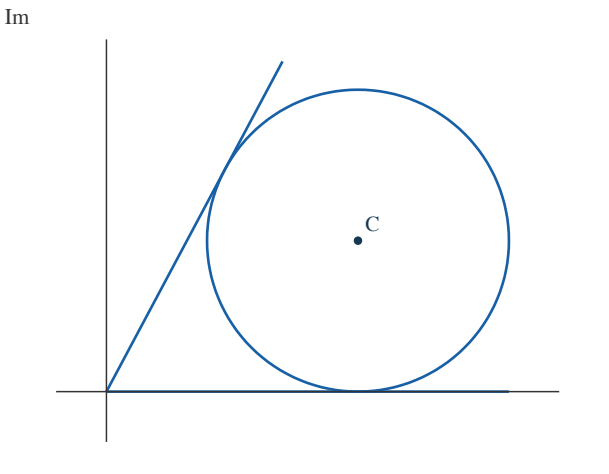
Sketch

Greatest argument and extreme moduli 27

For $|z-5-3i|=3$, find the greatest argument and the least and greatest modulus.

Argument

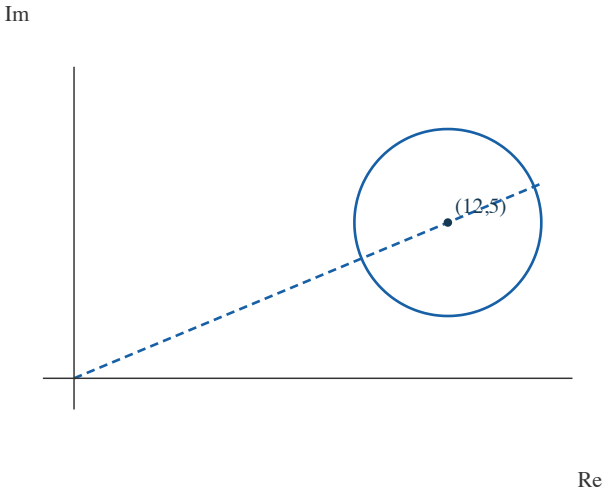
Modulus



Quickfire Test Your Understanding 28

Find the least and greatest values of $|z|$ on $|z-12-5i|=3$.

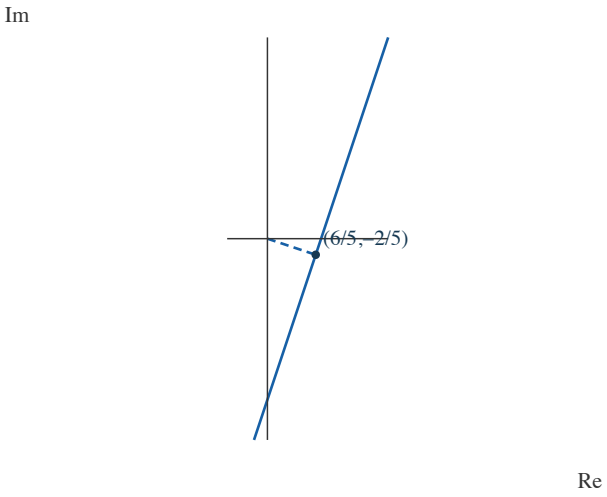
Distances



Least modulus on a line 29

Find the least modulus on the line $|z-3| = |z+i|$.

Minimum



Rays with a fixed argument 30

1. Sketch $\arg z = \pi/6$.
2. Sketch $\arg(z+3+2i) = 3\pi/4$.

1

2

Sketch

Sketch

4. For $|z - 4 - 3i| = 8$:

a. Give its Cartesian equation.

b. Sketch the locus.

c. Find the greatest and least modulus.

a

c

Sketch

5. Sketch $|z + 2 - 2\sqrt{3}i| = 2$. Find the smallest and largest principal argument.

Arguments

Sketch

7. Sketch $|z-3| = |z-6i|$, and find the least modulus.

Locus and distance

Sketch

8. Sketch $|z+3+3i| = |z-9-5i|$, and find the least modulus.

Locus and distance

Sketch

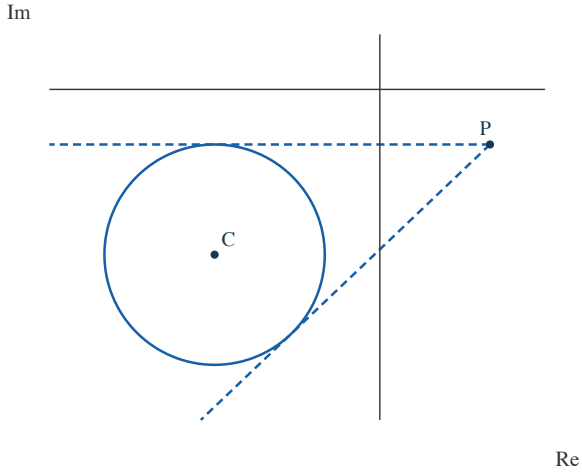
12. For $|z + 6 + 6i| = 4$:

a. Find the least and greatest modulus.

b. Find $-\pi < \theta < \pi$ for which the ray $\arg(z - 4 + 2i) = \theta$ has no point on the circle.

a

b



Regions 36

Shade the region satisfying all three conditions:

$|z - 4 - 2i| \leq 2$

$|z - 4| < |z - 6|$

$0 \leq \arg(z - 2 - 2i) \leq \pi/4$

Region

Sketch

Test Your Understanding 37

Shade

$|z-1| \leq 1, \quad \pi/12 \leq \arg(z+1) \leq \pi/2$

Region

Sketch

Exercise 2F Pearson Core Pure 1, p. 38 38

2. Shade $|z| \leq 5$ and $|z| \leq |z-6i|$.

2

Sketch

3. Shade $|z+1-i| \leq 1$ and $0 \leq \arg z \leq 3\pi/4$.

3

Sketch

4. Shade $|z| \leq 3$ and $\pi/4 \leq \arg(z+3) \leq \pi$.

Region

Sketch

5a. Sketch:

i. $|z-2| = |z-6-8i|$

ii. $\arg(z-4-2i) = 0$

iii. $\arg(z-4-2i) = \pi/2$

b. Shade $|z-2| \leq |z-6-8i|$ with $0 \leq \arg(z-4-2i) \leq \pi/2$.

a

b

Sketch

6a. Sketch $|z+10| = |z-6-4\sqrt{2}i|$ and $|z+1| = 3$.

b. Find their intersections.

c. Shade $|z+1| \leq 3$ with $|z+10| \leq |z-6-4\sqrt{2}i|$.

a

b

Sketch